ShopEase User Management Service

API Demo Evidence & Test Results

A ten-page report documenting profile and address API behavior, Swagger UI, authorization outcomes, and persisted demo records from a local run.

19 HTTP interactions	**9 / 9** expected status checks	**1** default address

Evidence boundary

The service was exercised over HTTP on a local FastAPI server using an isolated SQLite demo database and locally signed synthetic JWTs. This is a functional demo, not a production deployment. It does **not** represent live MySQL, a real Authentication Service, or Postman execution. All names, phone numbers, addresses, and IDs are fabricated for this report.

Captured on 11 November 2026 (UTC timestamps are embedded in the run data).

1. Service scope and API map

The service owns customer profile and address data. Authentication remains external; the profile owner is resolved from the verified JWT claim, not from client-supplied ownership fields.

Profile operations

Method	Endpoint	Purpose
POST / GET / PUT / PATCH / DELETE	/api/users/profile	Create, read, replace, partially update, or remove the current user profile
POST	/api/users/profile/deactivate	Deactivate the local profile only; does not revoke Auth Service credentials

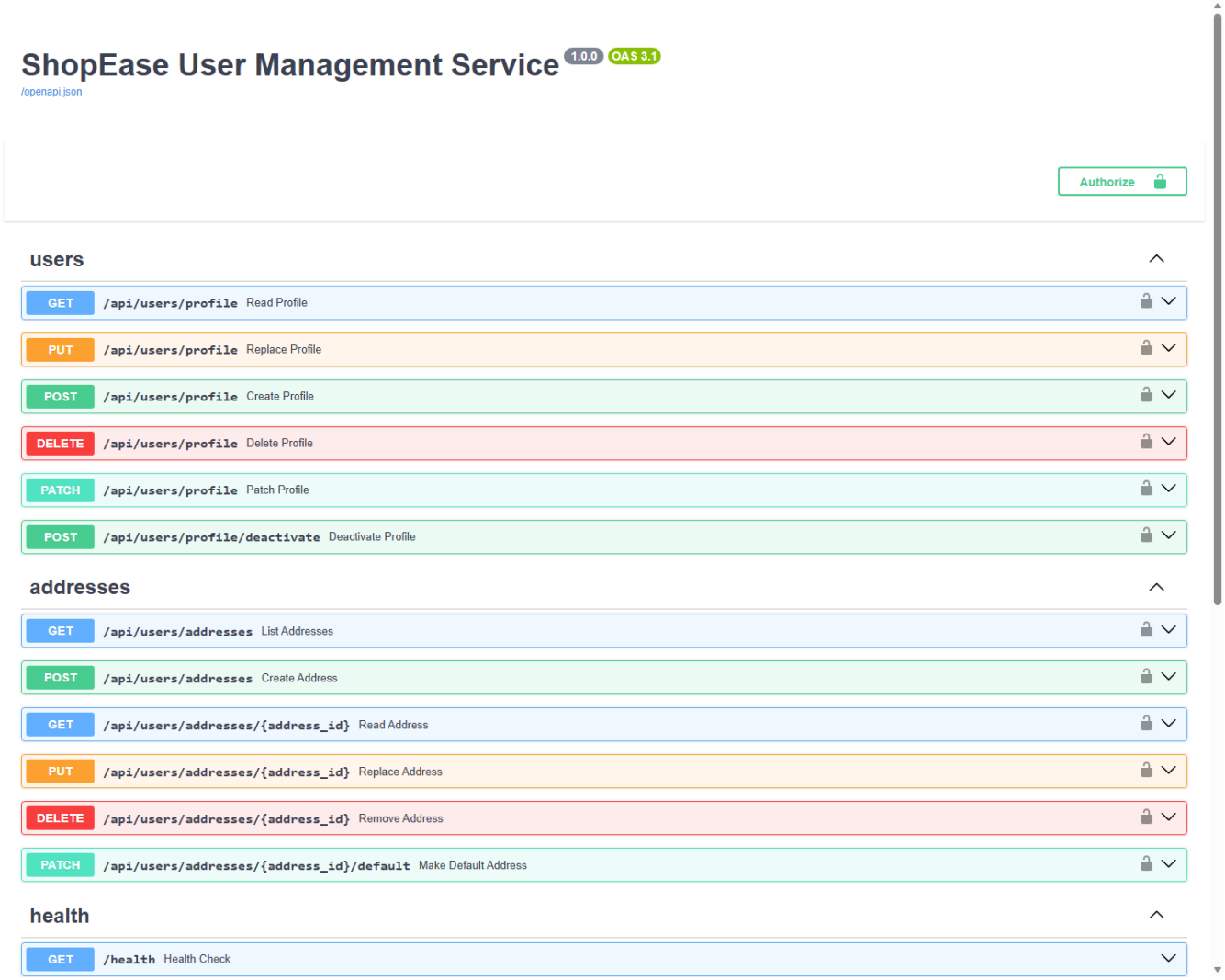
Address operations

Method	Endpoint	Purpose
POST / GET	/api/users/addresses	Create/list addresses for current profile
GET / PUT / DELETE	/api/users/addresses/{address_id}	Read/update/remove an address after ownership check
PATCH	/api/users/addresses/{address_id}/default	Set the current user?s default address

API docs: `/docs` ? Health endpoint: `/health`.

2. Swagger UI ? running service

Captured from the local service at `http://127.0.0.1:8765/docs`. The screenshot shows the documented profile, address, and health routes. Tokens are not shown in the screenshot.



Swagger UI documentation view captured from the running local FastAPI process.

3. HTTP scenario results

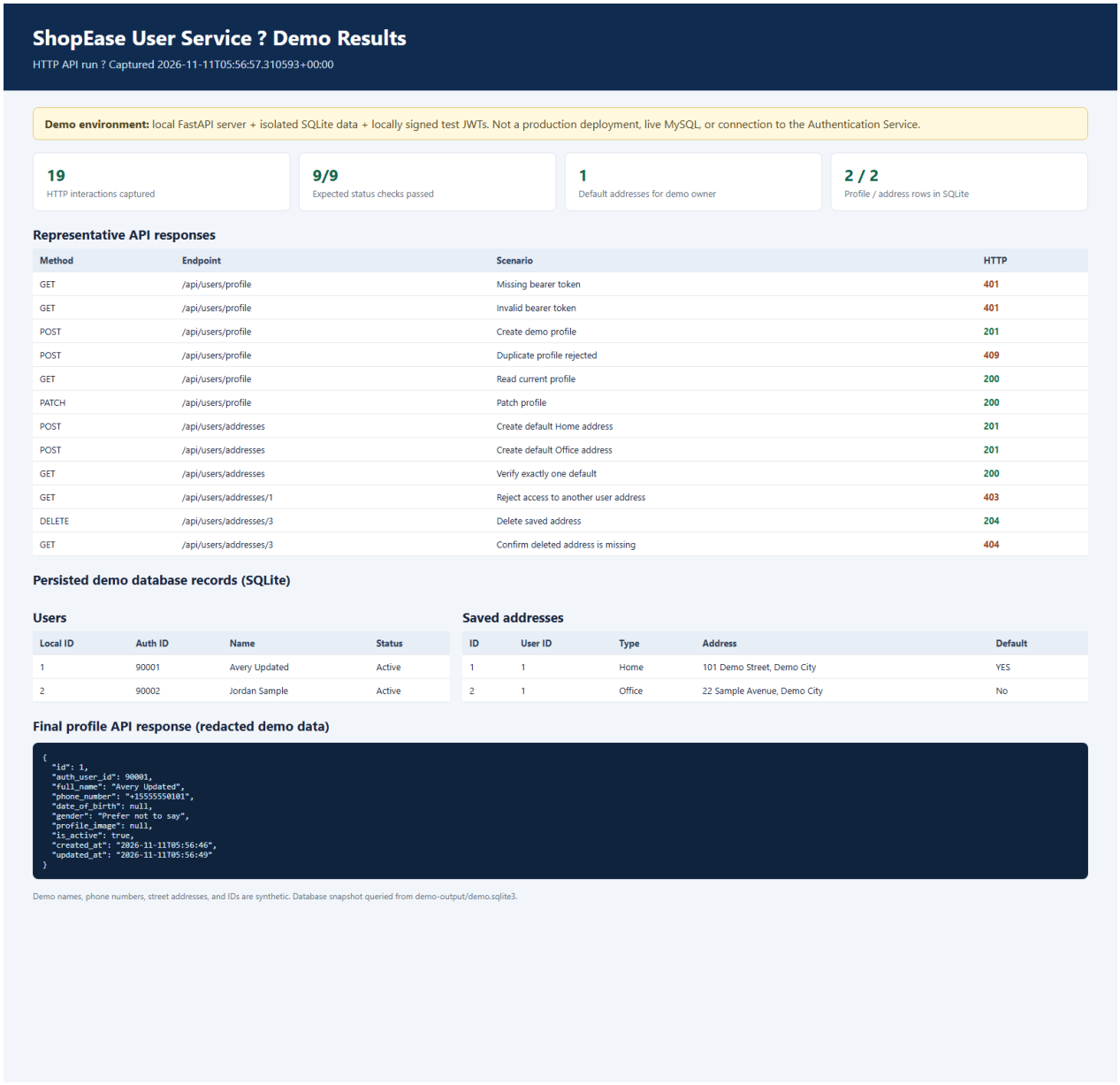
Representative API calls and observed status codes. Expected outcomes include authentication failures, duplicate prevention, access isolation, and deletion behavior.

Scenario	Method / path	HTTP
Missing bearer token	GET /api/users/profile	401
Invalid bearer token	GET /api/users/profile	401
Create demo profile	POST /api/users/profile	201
Duplicate profile rejected	POST /api/users/profile	409
Read current profile	GET /api/users/profile	200
Patch profile	PATCH /api/users/profile	200
Create default Home address	POST /api/users/addresses	201
Create default Office address	POST /api/users/addresses	201
Verify exactly one default	GET /api/users/addresses	200
Reject access to another user address	GET /api/users/addresses/1	403
Delete saved address	DELETE /api/users/addresses/3	204
Confirm deleted address is missing	GET /api/users/addresses/3	404

All listed status codes matched the expected result for this local demo. The full sequence captured 19 HTTP interactions; 9 status assertions were checked explicitly.

4. Demo response and database screenshot

The dashboard below is rendered from actual HTTP responses and a direct query of the demo SQLite database after the request sequence.



ShopEase User Service ? Demo Results

HTTP API run ? Captured 2026-11-11T05:56:57.310593+00:00

Demo environment: local FastAPI server + isolated SQLite data + locally signed test JWTs. Not a production deployment, live MySQL, or connection to the Authentication Service.

19	9/9	1	2 / 2
HTTP interactions captured	Expected status checks passed	Default addresses for demo owner	Profile / address rows in SQLite

Representative API responses

Method	Endpoint	Scenario	HTTP
GET	/api/users/profile	Missing bearer token	401
GET	/api/users/profile	Invalid bearer token	401
POST	/api/users/profile	Create demo profile	201
POST	/api/users/profile	Duplicate profile rejected	409
GET	/api/users/profile	Read current profile	200
PATCH	/api/users/profile	Patch profile	200
POST	/api/users/addresses	Create default Home address	201
POST	/api/users/addresses	Create default Office address	201
GET	/api/users/addresses	Verify exactly one default	200
GET	/api/users/addresses/1	Reject access to another user address	403
DELETE	/api/users/addresses/3	Delete saved address	204
GET	/api/users/addresses/3	Confirm deleted address is missing	404

Persisted demo database records (SQLite)

Users

Local ID	Auth ID	Name	Status
1	90001	Avery Updated	Active
2	90002	Jordan Sample	Active

Saved addresses

ID	User ID	Type	Address	Default
1	1	Home	101 Demo Street, Demo City	YES
2	1	Office	22 Sample Avenue, Demo City	No

Final profile API response (redacted demo data)


```
{
  "id": 1,
  "auth_user_id": 90001,
  "full_name": "Avery Updated",
  "phone_number": "+15555550101",
  "date_of_birth": null,
  "gender": "Prefer not to say",
  "profile_image": null,
  "is_active": true,
  "created_at": "2026-11-11T05:56:46",
  "updated_at": "2026-11-11T05:56:49"
}
```


Demo names, phone numbers, street addresses, and IDs are synthetic. Database snapshot queried from demo-output/demo.sqlite3.

The image displays response statuses, final profile and address rows, and the one-default-address invariant.

5. Profile API output

The demo creates a profile using the authenticated subject, rejects a second create for the same subject, and exercises profile read, full update, and patch operations.

Operation	HTTP	Observed result
Create profile	201	Profile created for Auth user 90001
Duplicate create	409	Duplicate profile prevented
Read profile	200	Current user profile returned
PUT profile	200	Profile fields replaced
PATCH profile	200	Gender field updated; remaining fields retained

Final profile response


```
{
  "id": 1,
  "auth_user_id": 90001,
  "full_name": "Avery Updated",
  "phone_number": "+15555550101",
  "date_of_birth": null,
  "gender": "Prefer not to say",
  "profile_image": null,
  "is_active": true,
  "created_at": "2026-11-11T05:56:46",
  "updated_at": "2026-11-11T05:56:49"
}
```


The request body did not supply an owner ID. The Auth Service user ID was derived from the verified demo JWT claim.

6. Address management and default selection

Two delivery addresses were created; both were initially sent as default. The API then set Home as default and the subsequent list response contained exactly one default record.

ID	Type	Address	Default
1	Home	101 Demo Street, Demo City	True
2	Office	22 Sample Avenue, Demo City	False

Observed address operations

Operation	HTTP	Result
Create Home default	201	Address 1 created
Create Office default	201	Address 2 created; prior default cleared
Set Home as default	200	Office changed to non-default
List addresses	200	2 saved addresses; one default
Delete temporary address	204	Address deleted; follow-up GET returned 404

Persisted final data: two active profiles and two retained addresses for the primary demo profile.

7. Authentication and ownership checks

Every protected request requires a bearer token. This demo used locally signed, synthetic tokens with the same configured claim format; no Auth Service was contacted.

Check	HTTP	Interpretation
No Authorization header	401	Unauthenticated request rejected
Malformed bearer token	401	Invalid token rejected
Second create for same owner	409	Uniqueness prevents duplicate profile
User 90002 reads user 90001 address	403	Cross-user address access denied
Deleted owned address read	404	Missing resource reported

Ownership model

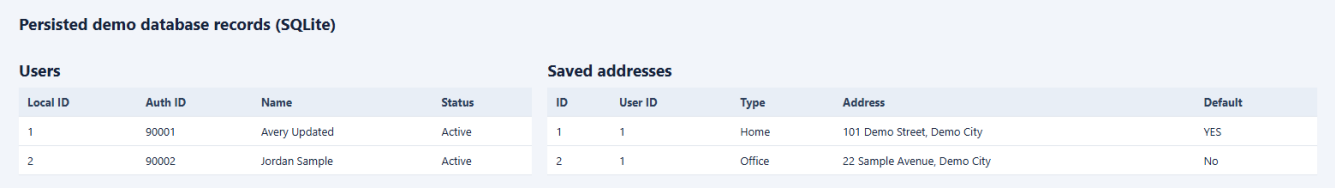
The profile is looked up by the authenticated Auth Service user ID. Address operations first resolve that user?s local profile and then verify the address?s local `user_id`. The demo confirms an address ID alone does not grant access.

Account-status boundary

Local profile deactivation is a User Service operation. It does not silently deactivate the Auth Service account or invalidate credentials there; the Authentication Service must separately disable the account and revoke or expire tokens.

8. Persisted demo database snapshot

The screenshot shows rows queried after the API sequence from the temporary SQLite database used to isolate the demonstration. It is not a MySQL schema verification.



Persisted demo database records (SQLite)

Users

Local ID	Auth ID	Name	Status
1	90001	Avery Updated	Active
2	90002	Jordan Sample	Active

Saved addresses

ID	User ID	Type	Address	Default
1	1	Home	101 Demo Street, Demo City	YES
2	1	Office	22 Sample Avenue, Demo City	No

Recorded state

Entity	Rows	Verification
users	2	Auth IDs 90001 and 90002 each have one local profile
user_addresses	2	Two addresses remain for local user 1
Default address count	1	Exactly one default for Auth user 90001

Schema responsibilities

The service model/migration defines a unique `auth_user_id`, a user-to-address relationship with cascade delete, and profile/address timestamps. This report validates demo rows, not migration application to a real MySQL server.

9. Validation summary and next steps

Completed in this evidence run

? Captured the actual local Swagger UI route list.
? Exercised 19 HTTP interactions through the running FastAPI app.
? Checked 9 expected status codes, including 401, 403, 404, 409, 201, 200, and 204 outcomes.
? Verified the default-address invariant after switching the default.
? Queried the final SQLite records and captured the result image.

Not performed / not claimed

? No live MySQL server or Docker Compose stack was started for this report.
? No real Authentication Service token was used; demo JWTs were signed locally for testing.
? No Postman run or production deployment was performed.
? This evidence does not replace integration testing against the target Auth Service and MySQL configuration.

To repeat against a configured environment

Start the service with its configured database, apply `alembic upgrade head`, obtain a real access token from the Authentication Service, and repeat the documented scenarios at `/docs` or in Postman. Verify database records and cross-user restrictions before release.

Files delivered: `ShopEase_User_Service_Demo_Report.pdf`, `swagger-docs.png`, `api-demo-results.png`, `database-snapshot.png`.